SIMATS ENGINEERING

ITA 06 – MACHINE LEARNING | Assessment Tool 1 (CO1) – Annexure A

Application Based Questions – Answers to Questions 1 to 5

Q1. Three boxes (Box B twice as likely) – Blue ball drawn, P(Box C)? (5 Marks)

1. Concept (Bayes' Theorem)

Bayes' theorem updates the prior probability of a box after observing the evidence (blue ball): P(C|Blue) = P(Blue|C)·P(C) / Σ P(Blue|Box)·P(Box).

2. Given data and priors

Box	Red	Blue	Green	Total	P(Box)	P(Blue|Box)
A	6	4	2	12	1/4	4/12 = 1/3
B	5	7	3	15	2/4 = 1/2	7/15
C	8	2	5	15	1/4	2/15

3. Calculation

- P(A)·P(Blue|A) = 1/4 × 1/3 = 1/12
- P(B)·P(Blue|B) = 1/2 × 7/15 = 7/30
- P(C)·P(Blue|C) = 1/4 × 2/15 = 1/30
- P(Blue) = 1/12 + 7/30 + 1/30 = 5/60 + 14/60 + 2/60 = 21/60 = 7/20

P(C|Blue) = (1/30) / (7/20) = 20/210 = 2/21

4. Final Answer

P(Box C | Blue) = 2/21 ≈ 0.0952 (≈ 9.52%)

Interpretation: Box C has very few blue balls and a low prior, so it is the least likely source of the blue ball.

Q2. Parking sections (Section B twice as likely) – Blue car, P(Section C)? (5 Marks)

1. Concept (Bayes' Theorem)

The same Bayes' rule is used: the probability of a section given the evidence (blue car) = (likelihood × prior) / total probability of blue.

2. Given data and priors

Section	Red	Blue	Black	Total	P(Sec)	P(Blue|Sec)
A	5	3	2	10	1/4	3/10
B	4	6	5	15	1/2	6/15 = 2/5
C	7	2	6	15	1/4	2/15

3. Calculation

- A: 1/4 × 3/10 = 3/40 = 9/120
- B: 1/2 × 2/5 = 1/5 = 24/120
- C: 1/4 × 2/15 = 1/30 = 4/120
- P(Blue) = (9 + 24 + 4)/120 = 37/120

P(C|Blue) = (4/120) / (37/120) = 4/37

4. Final Answer

P(Section C | Blue) = 4/37 ≈ 0.108 (≈ 10.81%)

Interpretation: Section B contributes most of the blue cars (24/37), so Section C is unlikely.

Q3. Breast cancer classification (Deep Learning + KNN) (5 Marks)

1. Concept

The confusion matrix (TP, TN, FP, FN) is used to evaluate a binary classifier. Here Positive = Malignant, Negative = Benign.

2. Finding the test set size and missing FN

- Total samples = 1800 × 2 = 3600
- Test data = 20% of 3600 = 720 samples
- FN = Total − (TP + TN + FP) = 720 − (210 + 190 + 30) = 720 − 430 = 290

3. Calculation

Metric	Formula	Calculation	Result
Accuracy	(TP+TN)/Total	(210+190)/720 = 400/720	55.56%
Precision	TP/(TP+FP)	210/(210+30) = 210/240	87.50%
Sensitivity (Recall)	TP/(TP+FN)	210/(210+290) = 210/500	42.00%
Specificity	TN/(TN+FP)	190/(190+30) = 190/220	86.36%

4. Explanation of effectiveness in medical diagnosis

- Precision (87.5%) and specificity (86.4%) are high: when the model says malignant it is usually right, and benign cases are rarely misclassified.
- Sensitivity is only 42%: the model misses 290 malignant cases (58% of cancer patients) as False Negatives.
- In cancer detection a False Negative is far more dangerous than a False Positive, since a missed cancer delays treatment.

Conclusion: Overall accuracy (55.56%) and low recall make the classifier not effective/reliable for screening in its current form; it needs improvement (more training, balanced threshold tuning, better features) to raise recall.

Q4. Training vs Validation Loss – Overfitting analysis (5 Marks)

1. Concept

Overfitting occurs when training loss keeps decreasing but validation loss starts to increase, i.e. the model memorises the training data instead of learning general patterns.

2. Data observation

Epoch	1	2	3	4	5	6	7	8
Train Loss	2.8	2.5	2.2	2.0	1.8	1.6	1.5	1.3
Val Loss	2.4	2.2	2.0	2.1	2.3	2.6	2.9	3.2

i) Epoch at which overfitting begins

Epoch 4. Validation loss is lowest at Epoch 3 (2.0) – the best model. From Epoch 4 the validation loss rises (2.0 → 2.1 → 2.3 → … → 3.2) while the training loss keeps falling (2.0 → … → 1.3). The gap (validation − training) widens from 0.2 at Epoch 3 to 1.9 at Epoch 8.

ii) Two techniques to reduce overfitting

- **Early stopping:** stop training at Epoch 3 when validation loss is minimum and restore those weights.
- **Dropout / L2 regularisation:** dropout randomly deactivates neurons and L2 penalises large weights, so the model cannot memorise the training data. (Data augmentation or more training data also helps.)

iii) Effect of overfitting on generalisation

An overfitted model gives very low error on training data but high error on unseen data. It learns noise and specific details of the training set, so its predictions on new data are unreliable. The growing gap between training and validation loss shows poor generalisation.

Q5. Training vs Validation Accuracy – Overfitting analysis (5 Marks)

1. Concept

When training accuracy keeps increasing but validation accuracy stops improving and then falls, the model has started to overfit.

2. Data observation

Epoch	1	2	3	4	5	6	7	8
Train Acc (%)	60	65	70	75	80	85	88	92
Val Acc (%)	58	63	68	72	73	72	70	68

i) Epoch at which overfitting starts

Epoch 6. Validation accuracy peaks at Epoch 5 (73%) – the best generalisation point. At Epoch 6 it drops to 72% while training accuracy rises to 85%, and it keeps falling (70%, 68%) while training reaches 92%. The gap grows from 7% (Epoch 5) to 24% (Epoch 8).

ii) Why validation accuracy decreases while training accuracy increases

The model begins to memorise training examples, including noise and outliers, instead of learning general patterns. This improves training accuracy but the memorised details do not apply to unseen validation data, so validation accuracy falls.

iii) Two methods to improve validation performance

- **Early stopping:** stop at Epoch 5, where validation accuracy is highest.
- **Regularisation (Dropout/L2) and data augmentation:** reduce model complexity and increase data variety, so the model generalises better.

Rubric Coverage Summary (Calculation Rubrics)

Rubric Criteria	Weightage	How it is addressed in the answers
Conceptual Understanding	25	Concept stated first in every answer (Bayes' theorem, confusion matrix, overfitting)
Accuracy of Answer	25	Correct final values: 2/21, 4/37, 55.56/87.5/42/86.36 %, Epoch 4, Epoch 6
Application of Knowledge	20	Formulas applied step by step to the given data
Completeness of Response	15	All parts (i, ii, iii) of every question answered
Clarity & Presentation	15	Tables, steps and bold final answers